CAPSTONE PROJECT CONTEXT & ARCHITECTURE

Universal Multi-Brand Physiological Strain, Contributor Attribution & Recovery Intelligence Engine

A detailed technical specification for a wearable-agnostic wellness engine that learns personal baselines, distinguishes activity-associated elevation from other physiological deviations, summarizes recovery context, and produces evidence-linked, conservative guidance.

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Scope Single universal product; research modules are evidence-gated

DESIGN PRINCIPLE / Keep the signal honest. Personalize the reference. Explain the evidence. Do not invent causes.

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01 Executive summary

The capstone is a single universal multi-brand wearable product: a capability-aware engine that ingests whatever legitimate health and movement records a connected device provides, normalizes them into versioned contracts, estimates within-person physiological deviation, separates co-occurring contributor evidence, summarizes recovery context, and returns conservative wellness guidance with explicit confidence and data-quality provenance.

The central engineering challenge is not to make all devices look identical. It is to preserve the real fidelity of each source while giving the product one stable set of meanings. A device with raw tri-axis accelerometry can support movement-intensity features that a summary-only device cannot. A device that reports nightly HRV can support a daily trend, but not an invented second-by-second HRV stream. The engine must adapt its resolution and confidence to the data actually available.

The core contribution is a combination of four research hypotheses:

Novelty	Core idea	Evidence needed
1. Personal robust baseline	Compare a person with their own eligible historical distribution, not only global fixed thresholds.	Causal rolling baseline; adequate history/coverage; comparison against global thresholds; no future leakage.
2. Context-aware contributor attribution	Separate activity-associated elevation, non-exertional physiological elevation, recovery-context deviation and optional vital-trend flags. Multiple contributors may coexist.	Compatible targets, activity/context controls, participant-held-out tests, calibrated confidence and honest interpretation.
3. Recovery trajectory and guidance	Use longitudinal signal changes and observed recovery episodes; associate policy guidance with evidence.	Define event onset/end, require valid episodes, evaluate held-out trajectory error/uncertainty before predicting time-to-recovery.
4. Universal capability-aware orchestration	Integrate brands/sources through capability declarations, canonical signal contracts, quality and graceful fallbacks.	Demonstrate one real adapter end-to-end; test missing channels, different cadences, data quality and per-source behavior.

> Critical product promise
> The engine reports measured physiological patterns and contextual associations. It does not diagnose illness, infer a specific external cause, or prove that an elevated heart rate means psychological stress. Model explanation is not causal inference.

The live product should begin with robust personal baselines, quality-aware engineered features, activity/context gates, and a supervised association model only where its labels genuinely support that target. More complex heads, forecasting, NormWear-derived accelerometer embeddings, sleep-stage fallback and LLM narration are separate evidence-gated capabilities. The product remains one system; it does not become a collection of dataset demos.

02 Project context, scope and research framing

2.1 Problem statement

Commercial wearables expose heterogeneous signals, summaries and proprietary scores. Different brands provide different sensors, sampling rates, timing semantics and access paths. Many devices show heart rate, sleep and activity, yet the interpretation of any one measurement depends on the user’s own reference, recent activity, measurement quality and history. A universal strain/recovery engine must therefore solve data harmonization, personalization, contributor evidence, uncertainty and explanation together.

2.2 Product objective

Given a time-stamped history from one or more supported data sources, estimate the user’s current physiological deviation relative to an eligible personal reference, summarize likely measured contributors and contextual associations, compute a recovery-context view from valid recent HRV/sleep/resting-HR/activity data, and provide conservative guidance. Every output must expose which sources and windows support it, what is missing, and how much temporal resolution the inputs permit.

2.3 In scope

- One product with a canonical signal contract and a capability registry for different devices, APIs and exported datasets.
- Personal baselines for HR, valid HRV metrics, sleep, activity/load and optional vital trends when available.
- Raw accelerometer features when actual axis data, units and gravity semantics are known; lower-resolution activity proxies otherwise.
- Activity/exertion evidence, non-exertional physiological elevation associations, recovery-context deviations, and cautious respiratory/oxygen/temperature trend flags.
- Multi-label/evidence fusion; contributor-level uncertainty and data coverage; model-versioned outputs.
- Observed recovery trajectories first; predicted recovery duration only after endpoint definitions and held-out validation.
- Rule-based wellness recommendations, optionally narrated by an LLM using only validated structured outputs.

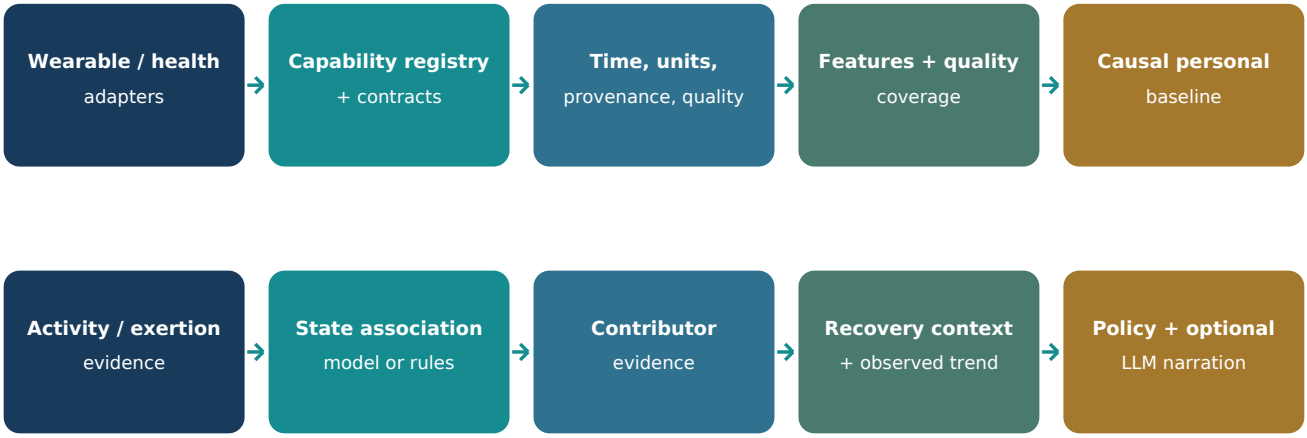
2.4 Explicitly out of scope

- Clinical diagnosis, disease prediction, or replacing clinician advice.
- Claiming a direct measurement of psychological stress from HR or HRV alone.
- Asserting an external or causal reason for a physiological reading from correlational wearables data.
- Inventing sensor channels, backfilling missing values with zeros, or converting day summaries into fake high-frequency waveforms.
- Using WESAD-specific trained heads or BVP/EDA/TEMP data paths as the universal product model.
- Claiming universal vendor compatibility before a real adapter and access path have been verified.

The research framing is intentionally conservative: the system is an estimator of personal physiological deviation and context, not a truth oracle for internal emotion or external cause.

03 System architecture at a glance

The universal pipeline has one flow from source adapters to a structured output. Raw accelerometer devices and summary-only devices enter through different adapters but converge on shared contracts and downstream semantics.



The same engine adapts to each source; unavailable signals remain missing, not fabricated.

Figure 1. End-to-end architecture. Adapter diversity is handled at ingestion; data quality and temporal resolution remain visible through every downstream stage.

3.1 Architectural layers

Layer	Responsibilities	Primary artifacts
1. Source/adapters	Ingest native or exported records; identify device/model/SDK and actual capabilities.	Adapter output, capability declarations, consent/access metadata.
2. Contracts + quality	Validate units, timestamps, provenance, duplicate records, cadence, quality and coverage.	Canonical observations, validity flags, missing reasons, quality report.
3. Feature engineering	Compute movement, cardiovascular, sleep, daily-context and temporal features only when inputs support them.	Versioned feature records with feature window and source lineage.
4. Personal baseline	Maintain causal robust reference distributions with minimum sample and coverage gates.	Baseline statistics, data sufficiency state, robust deviation values.
5. State/contributor estimation	Use rules and compatible trained heads; preserve multiple co-occurring contributor signals.	State associations, contributor evidence, confidence and calibration metadata.
6. Recovery/guidance	Summarize recovery context and observed trends; map valid patterns to conservative policy.	Recovery context, observed trajectory, recommendation policy IDs.
7. Product/API	Deliver transparent summary, supported time scale, data completeness and per-signal provenance.	Versioned JSON response, dashboard cards, daily report.

3.2 Capability tiers are fidelity tiers, not separate products

Tier	Commonly available inputs	Supported behavior and limitation
Motion-rich	Raw three-axis ACC; HR; possibly IBI; sleep/workout/activity summaries; optional vitals.	Movement intensity, ENMO and activity bouts where sampling and units permit; HR/activity alignment; valid HRV only from actual suitable data.
Standard health-data	HR/HRV samples or summaries; sleep; steps; workouts; optional SpO2/respiration/temp.	Context and trend calculations at native cadence; no raw-waveform claims unless exported raw data truly exist.
Summary-only	Resting HR, nightly HRV, sleep totals, daily steps/energy and workout summaries.	Daily baseline and recovery context only. No minute-level exertion classification or raw-ACC features.

Resolution rule: Output frequency and confidence never exceed what the required source observations support. Missing fields, coarser summaries and reduced confidence remain explicit.

04 Inputs and universal signal contract

Every data value enters as an observation with source and timing metadata. Normalized units do not erase native units, source cadence or signal provenance. Canonicalization creates comparability; it does not turn distinct measurement methods into an identical sensor.

4.1 Supported signal families

Signal family	Preferred inputs	Allowed use	Important limits
Heart rate	Timestamped HR series; resting HR; device summaries.	Personal deviation; HR/activity alignment; recovery context.	HR alone is not a stress ground truth. Preserve source cadence and aggregate windows.
HRV / IBI	Device-reported metric with window metadata; actual timestamped inter-beat intervals when exposed.	Within-person nightly/valid-window HRV deviation; HRV trend.	Never derive RMSSD/SDNN from rounded or smoothed BPM. Preserve RMSSD and SDNN as distinct metrics.
Raw movement	Three-axis accelerometer with units, sampling rate and sensor semantics.	ENMO, movement intensity, rolling statistics and activity bouts.	No assumption every platform exposes raw ACC; verify axis orientation, scale and gravity semantics.
Activity summaries	Steps, workout records, active energy/calories, duration and activity type.	Lower-resolution activity proxy and load context.	Do not equate estimated calories to metabolic ground truth or fabricate ENMO/NormWear embeddings.
Sleep	Duration, stages, efficiency, sleep/wake intervals if provider exposes them.	Sleep/recovery context and nightly trends.	Never infer stages from HR alone; report missing stage data as missing.
Optional vitals	SpO2, respiratory rate, skin-temperature deviation or temperature where supported.	Additional personal trend flags and context.	Not stress labels or diagnoses; quality/freshness and provider limitations apply.

4.2 Canonical observation envelope

```
Observation {
  participant_id: pseudonymous identifier
  signal_name: canonical signal enum
  value: scalar | vector | structured measurement
  canonical_unit: unit after documented conversion
  native_unit: source-reported unit
  event_start_utc / event_end_utc: nullable timestamps
  observed_at_utc: timestamp if source defines one
  user_timezone: IANA timezone for local-day aggregation
  cadence_hz: nullable; aggregation_interval: nullable
  modality: raw | derived | device_summary | self_report
  source: adapter_id, manufacturer, model, SDK/API, version
  quality: validity, flags, coverage, artifact state, freshness
  missing_reason: unsupported | permission | not_recorded | invalid | stale
  schema_version, provenance_id, import_time, source_record_id
}
```

The exact schema should be represented as typed Python/Pydantic models or an equivalent runtime-validated contract. Schema changes require a version bump and contract tests.

4.3 Timestamp and alignment policy

- Convert actual timestamps to UTC internally. Preserve the user’s timezone for local-day summaries and daylight-saving boundary logic.
- Do not reinterpret session-relative elapsed seconds as real calendar dates. Mark dataset-relative timelines explicitly.
- Align observations into a feature window only when event timing and coverage thresholds are acceptable. Keep original timestamps and aggregation intervals.
- Deduplicate using stable source identifiers where available and conservative time/value rules otherwise; record what was deduplicated.
- Never interpolate nightly/daily summaries into continuous streams. Do not assume streams from different sensors are synchronized.

- Preserve native sample cadence. Any resampling requires a declared target rate, appropriate anti-aliasing/quality treatment, and evidence that the encoder supports it.

4.4 Missingness is a first-class state

A missing SpO2 field is not zero SpO2; absent raw ACC is not zero movement; absent HRV is not normal HRV. Use an explicit missing reason and a feature-availability indicator. A fallback must have different semantics: for example, steps are a lower-resolution activity proxy, never a synthetic ENMO signal or a synthetic NormWear vector.

> Minimum rule
> Never silently zero-fill unavailable physiological channels, and never label “no observed deviation” when the relevant measurement is absent, stale or too low quality.

05 Ingestion, data quality and preprocessing

5.1 Adapter and capability registry

Each adapter declares what it can actually return, at what cadence, under what access permissions and on which device/OS/API versions. Capabilities are specific to the data path, not just the consumer product name. A signal visible in a consumer app may be unavailable to third-party APIs or may be a daily summary rather than a raw stream.

Registry field	Purpose
source_id, adapter_version	Tie every record to a tested ingestion path and released adapter code.
manufacturer, model, OS/SDK/API version	Make compatibility claims narrow, auditable and version-specific.
available_signals + access_mode	Distinguish raw live stream, export, minute sample, nightly summary, daily aggregate and derived score.
native_units + cadence + window	Preserve measurement meaning and temporal resolution.
permission/consent requirements	Avoid assuming data access without explicit supported permission and user action.
known restrictions + quality profile	Make gaps, supported regions, model limitations and expected missingness visible.

Capability snapshot checked 15 October 2026: Samsung Health Sensor SDK documentation lists continuous raw tri-axis accelerometry at 25 Hz and raw PPG at 25 Hz; processed heart rate with IBI at 1 Hz; EDA continuous on Galaxy Watch8 series and later; and processed skin-temperature readings on supported Galaxy Watch models. Exact model, OS, SDK and foreground/background behavior matter, and skin temperature is not core body temperature. Google's current Health API documentation says the legacy Fitbit Web API is scheduled to turn off on 30 October 2026 and new project onboarding is not currently open. Treat these as time-sensitive access checks, not guarantees; re-verify official docs and obtain a working authorized path before claiming support. Samsung specifications; Google Health access and migration.

5.2 Quality gates

Gate	Checks	Failure behavior
Schema	Required keys, type, unit, schema version, enum validity.	Reject or quarantine malformed records; emit an explicit validation error.
Temporal	Ordering, duplicate/overlap, timezone, interval duration, stale samples.	Exclude affected windows; keep source record for audit if policy permits.
Physiological plausibility	Signal-specific range checks plus provider quality flags.	Flag suspicious/out-of-range value; do not “repair” it without a documented rule.
Coverage	Fraction of expected observations in a feature window and expected cadence.	Mark insufficient coverage; reduce confidence or suppress estimate.
Cross-signal alignment	Overlap of HR, ACC, IBI and event context over the same evidence window.	Do not claim coincidence if timestamps do not support it.
Baseline eligibility	Not stale; adequate valid samples; avoid active workout for resting baseline.	Do not update baseline; retain baseline confidence/status.

5.3 Feature windows

Use multiresolution windows matched to the signal and task. Raw ACC can support short windows (for example six seconds) and rolling summaries (for example five, fifteen and sixty minutes) when the sampling rate and data quality support them. HR series can have their own sampling cadence. Nightly HRV and sleep summaries are daily/nightly context. A feature should include its time window, source cadence, quality and aggregation operator.

5.4 Accelerometer processing and ENMO

When the accelerometer axes are expressed in g and gravity semantics are known, Euclidean Norm Minus One (ENMO) can be calculated as:

```
ENMO(t) = max(0, sqrt(ax(t)^2 + ay(t)^2 + az(t)^2) - 1 g)
```

If the axes are expressed in m/s^2, convert consistently or use an equivalent gravity constant in matching units. ENMO is a movement-intensity feature, not a universal energy expenditure measurement. Candidate rolling features include mean, variance, quantiles, signal-magnitude distribution, movement-bout duration, and active/resting context.

5.5 Sampling-rate caution

Do not blindly resample 25-Hz raw ACC to 64 Hz just to mimic an old six-second WESAD tensor. Interpolation creates additional points but does not recover missing bandwidth. Validate the target encoder on native or justifiably canonical sampling rates, preserve sampling-rate metadata, and document any anti-aliasing. A six-second window at 25 Hz contains 150 samples; the correct input contract depends on actual encoder support, not on matching an unrelated legacy shape.

06 Personalized baseline and temporal context

6.1 Role of the baseline

The baseline estimates what is typical for this user under a defined context. It is not a population norm, a clinical normal range, or a guarantee of wellness. Baselines should be feature-specific and context-aware: resting HR is not the same distribution as workout HR, nightly HRV is not the same measurement as a short active-window HRV estimate, and sleep duration is not interchangeable with sleep stage proportions.

6.2 Robust deviation

For a feature x and eligible prior personal observations, use a robust center and scale, such as the median and median absolute deviation (MAD):

```
robust_z = (x - median_prior) / (1.4826 * MAD_prior)
```

The factor 1.4826 is a common normal-consistency scaling for MAD. Apply feature-specific scale fallbacks, minimum sample counts, winsorization/clipping and data-coverage gates. A tiny epsilon alone does not solve zero-MAD behavior: if the user’s historical feature is nearly constant, the implementation needs a justified fallback scale or must mark the feature as insufficiently calibrated.

6.3 Baseline eligibility and update policy

- Use only measurements that occurred before the score being calculated; never include future observations in the baseline.
- Use a configurable 7-30 day calibration candidate, but require actual minimum observations, valid-night counts, coverage and temporal spread before declaring it reliable.
- Classify the initial baseline as provisional when history is sparse. A nominal three-day baseline should not be presented as a stable personal norm without supporting coverage.
- Exclude active workouts and poor-quality data from resting references. Use context tags or separate distributions where enough data exists.

- Update slowly and transparently, after quality checks, to avoid allowing a sustained abnormal or unusual period to redefine the person's norm too quickly.
- Expose baseline version, calibration period, valid observation count, last update and current confidence.

6.4 Baseline states

State	Interpretation	Product behavior
unavailable	No suitable history or no valid values for that feature.	Do not invent a personal deviation; use only supported context and show limitation.
provisional	Some valid history exists but data are sparse or coverage is uneven.	Produce conservative low-confidence context; do not overstate deviation.
calibrated	Minimum sample, coverage and freshness gates pass.	Use robust deviation and show the reference window/quality.
stale	A previously calibrated reference exists but current updates/data are too old.	Mark stale, limit confidence, request refresh or avoid current claims.

6.5 Temporal forecasting

A causal trend model or scalar time-series forecaster may be tested for a narrowly defined task such as short-horizon HR forecasting, but it is not automatically a stress classifier, a personal baseline, a contributor-attribution model or a recovery predictor. Benchmark against simple last-value, rolling-median and linear-trend baselines. Keep forecasting optional unless it demonstrably improves the target task on participant-held-out data.

Model outputs must follow the documented API. A scalar forecaster must not be described as a vector embedding unless that embedding path actually exists and has been validated.

07 Movement and cardiovascular model stack

7.1 Movement feature extractor

A product-compatible movement path has two modes that keep their semantics separate:

Mode	Input	Features/outputs	Limit
Raw-ACC mode	Real tri-axis accelerometer, units/gravity verified, cadence and quality preserved.	ENMO; vector magnitude; rolling mean/variance/quantiles; movement bouts; optionally a frozen NormWear embedding if a downstream held-out ablation helps.	No raw-ACC feature when the source lacks axes; device orientation, sampling rate and motion artifacts matter.
Summary-proxy mode	Steps, workout events, active-energy/calorie summaries and timestamps.	Activity/load proxy, workout recency/duration, daily steps/energy context.	Does not create ENMO, raw movement vectors or NormWear embeddings; temporal granularity remains lower.

7.2 NormWear encoder: optional, narrowly scoped

The existing NormWear checkpoint and its official continuous-wavelet-transform (CWT)/encoder implementation are relevant only as a possible frozen representation for real three-axis IMU/accelerometer input. The existing model is reported as a ViT-Base-scale encoder with 136.1 million parameters and a 768-dimensional embedding. Use its output as a candidate kinematic representation, not as an already-trained universal activity or stress classifier. The old WESAD downstream heads were trained for a different input representation and should not be reused as universal classifiers.

- Pin the official encoder implementation, checkpoint hash and license/version in an artifact manifest.
- Use the encoder only if raw ACC is truly available and normalization, channel order, sample rate and window duration are supported by the implementation.
- Compare ENMO/statistical features alone with NormWear embeddings alone and with the fused feature set; use participant-held-out evaluation.

- Fit PCA, if used, on training participants only and evaluate whether the reduced dimension preserves downstream performance. An eight-dimensional projection is a candidate, not a design law.
- If embeddings do not add measurable value, or available labels/data are inadequate, exclude NormWear from the live product without affecting the rest of the architecture.

A previously recorded checkpoint hash is `36d0bca18356ccfc8e8916058bf838f26f1212a646f5780b487ad78581a92561`; verify the exact file hash before relying on this value. Historical WESAD results are not proof of universal ACC performance.

Legacy WESAD results - archive only, not product claims

The old WESAD tensor contract used six-second windows with three-second stride at 64 Hz and channel order BVP / EDA / TEMP, commonly represented as [B, 3, 384]. Reported historical held-out results were: 768-D NormWear-only Model A ROC-AUC 0.9400 and F1 0.8113; 20-D feature Model B ROC-AUC 0.8977; 788-D fusion Model C ROC-AUC 0.8708. These figures have not been independently reproduced here and must not be presented as the new product's metrics. The old processed-window count also conflicted between 9,066 and 9,078. Rebuild a deterministic subject/window manifest before making any legacy benchmark claim. Do not retain the old six-second RMSSD/SDNN features.

7.3 Activity/exertion classifier

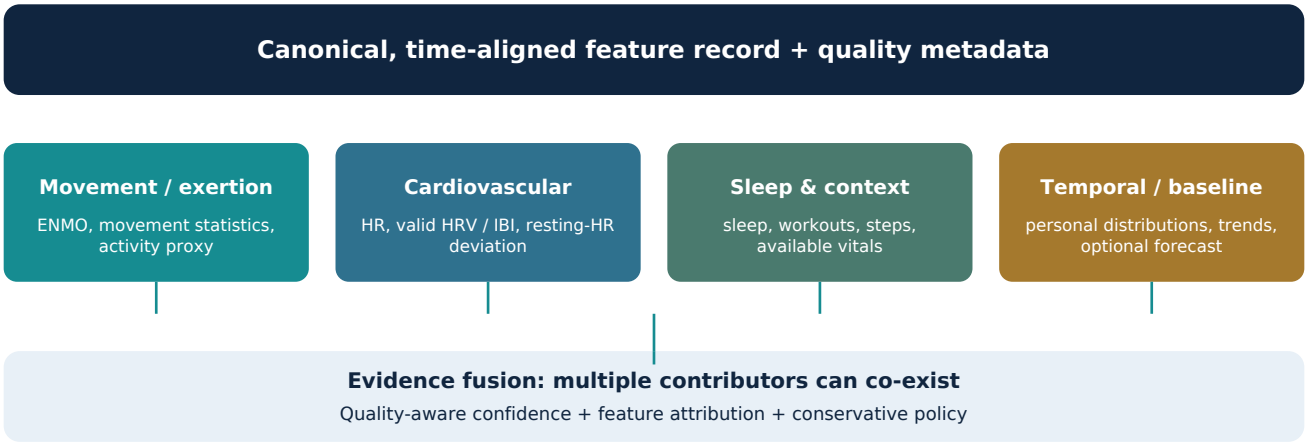
A supervised activity classifier may be trained on a compatible activity-labeled dataset (PAMAP2 has been considered in the broader research plan), but the output semantics are activity classes or activity/exertion context, not stress. Use participant-disjoint splits, compare a simple feature baseline, and test transfer to the target device/source. If that transfer cannot be supported, keep activity estimates at the transparent feature/rule level.

7.4 Cardiovascular feature builder

- Preserve HR at native cadence and compute context relative to personal baselines, including resting-HR deviation where the provider supports it.
- Accept device-reported HRV only with its metric type, measurement window, units and provenance; compare like with like.
- Compute RMSSD/SDNN only from actual, timestamped, quality-checked IBI/NN intervals over an explicitly defined valid window. Do not assume every reported interval is artifact-free.
- RMSSD and SDNN are distinct metrics and should remain separate features. Their suitable windows and error characteristics depend on sensor, objective and artifact handling.
- When only nightly HRV is available, use a nightly/daily context; do not interpolate it into a minute-level trace.

Why separate feature heads? Movement, cardiovascular context, sleep and optional vital trends have different physical meanings and label requirements. Separate builders allow independent validation and graceful disabling when a device lacks required inputs.

08 State estimation and contributor fusion



A model head is enabled only when its training target and validation support are real.

Figure 2. Specialist feature streams feed an evidence fusion layer. Multiple contributor states may co-occur; a single mutually exclusive softmax is not a good default for overlapping physical exertion, recovery deficit and vital-trend deviation.

8.1 State semantics

Output state	Operational definition	What it does not mean
activity_associated_elevation	HR or related physiology is elevated in a window that overlaps valid movement/workout evidence.	Does not prove exertion caused every physiological change; overlap/quality must support the association.
non_exertional_physiological_elevation	Physiological elevation relative to personal baseline with low measured movement and known recent-workout context.	Not synonymous with psychological stress, anxiety, illness or a specific external trigger.
recovery_context_deviation	Valid sleep, nightly HRV, resting-HR or recent-load context differs from the personal reference.	Not a diagnosis, validated readiness truth, or proof of accumulated fatigue.
respiratory_or_oxygen_deviation	Available respiratory-rate/SpO2/temperature feature differs from personal reference or crosses a cautious technical flag.	Not a diagnosis of infection, hypoxia or other medical state.
insufficient_data / ambiguous_context	Quality, coverage, timing or signal disagreement prevents a defensible estimate.	Not a normal state. It explicitly represents missing evidence.

8.2 Specialist-head design

Head	Candidate inputs	Target / supervision	Candidate method	Gate / caveat
Activity / exertion	ENMO/statistics; optional NormWear ACC embedding; steps/workouts proxy.	Labeled activity/exertion or activity context.	LightGBM/XGBoost or transparent rules.	Activity-label quality and transfer to wearable context must be checked.
Acute task-state	HR/IBI/ACC and event timing where aligned.	Task phase such as rest/stress/recovery in a laboratory dataset.	Tree model or temporal model if sample supports it.	Laboratory task phase is not a universal real-world psychological-stress label.
Self-report association	Valid IBI/ACC/HR and contemporaneous report timing.	Association with a user's self-reported state.	Calibrated XGBoost/LightGBM if labels support it.	Do not rename reported stress as objective or clinical stress.
Recovery context	Nightly HRV, sleep, resting HR, recent activity/load and deviations.	Initially rule-based/personalized descriptive index.	Deterministic aggregation; ML only with direct outcomes.	Self-defined score is not a physiological ground truth.
Vital-trend detector	SpO2/respiration/temperature only when measured.	Within-person deviation / quality-aware flag.	Robust anomaly detector or transparent threshold.	Context flag only; no diagnosis or cross-device raw-score comparability without calibration.
Recovery trajectory	Eligible event windows with observed pre-event baseline and recovery endpoint.	Time from event end/onset to explicit return-to-reference definition.	Simple exponential/quantile regression or survival/time-series model only if warranted.	Do not predict minutes-to-recovery without enough valid episodes and held-out evaluation.

8.3 Dataset label semantics must stay separate

Activity labels, lab task phases, user self-reports, nightly sleep diaries and physiological anomaly flags are not interchangeable labels. They should not all be concatenated into one "stress" target. If the product later uses a multi-head system, each head must keep its own target definition, training data lineage, calibration, model version and eligible-device/feature requirements.

8.4 Fusion rules

- Allow more than one contributor to be present at once; do not force exertion, poor sleep and abnormal vital trends into mutually exclusive classes.
- Use quality and missingness gates before fusion. A feature unavailable on one brand must not lower its numeric value to zero or be interpreted as reassuring.
- Emit evidence records that include signal, feature value, personal reference, deviation direction/magnitude, observation window and provenance.
- Keep state association, feature deviation and recommendation trigger distinct fields. A heuristic state is not a validated classifier probability.
- When conflict exists between heads or timestamps, return an ambiguous/low-confidence state rather than a fabricated single primary cause.

09 Feature contract and baseline-relative score

9.1 Avoid a fixed, unverified dimensional contract

A feature vector contract must be named, versioned, unit-tested and derived from actual code. Do not lock the system to a stale 28-dimensional vector while module descriptions disagree about whether the temporal representation has seven or eight values, or whether ENMO_5m is included. Define features by semantic name and metadata, then let schema validation enforce what a given model expects.

9.2 Suggested feature families

Family	Illustrative features	Metadata retained
Movement	ENMO mean/quantiles; vector-magnitude statistics; movement-bout duration; step/workout proxies.	Source mode raw vs summary; cadence; time window; wear/quality coverage.
Cardiac	HR median/quantiles; baseline-relative HR; resting-HR deviation; valid RMSSD; valid SDNN; IBI coverage.	Metric type; native window; units; artifacts/quality; sensor provenance.
Sleep/recovery	Sleep duration/efficiency/stage durations if exposed; nightly HRV; sleep regularity; recent workout context.	Night definition/timezone; source/device stage availability; missing stage state.
Temporal/context	Recent trend; prior activity; workout recency; time since sleep/wake where supported; repeated baseline deviations.	No future leakage; clock/timezone assumptions; aggregation window.
Optional vitals	SpO2 summary, respiratory-rate summary, skin-temperature deviation.	Device/source semantics; coverage; freshness; quality flags; availability state.

9.3 Personal physiological deviation index

A 0-100 index can be used as a product-facing composite, but it is an engineered relative index rather than a clinical unit. The index must be versioned and calibrated against an explicit outcome if any performance or comparability claim is made. When no reliable labelled outcome exists, call it an experimental personal-deviation index and display the evidence/coverage alongside it.

Do not retain fixed heuristic weights (for example 0.65/0.35) or unvalidated conversions from baseline z-scores into strain points merely because a previous prototype used them. Choose aggregation logic from the intended semantics, estimate/calibrate with supported data when possible, and document all assumptions.

9.4 Confidence and coverage

Confidence should describe evidence availability and model calibration, not the assistant’s tone. It can combine documented coverage, source quality/freshness, baseline sufficiency, model applicability and calibration quality. Do not multiply arbitrary component scores and call the result a probability. If a probability is presented, the model’s target, calibration method, validation split and reliability must be documented.

Output dimension	Example interpretation
Data coverage	Share of expected observations that passed quality gates in a defined window.
Baseline confidence	Unavailable, provisional, calibrated or stale, supported by counts and history.
Model confidence/calibration	Only for a trained model with an explicit target and calibration evaluation.
Temporal resolution	Instantaneous/sample-level, window-level, nightly or daily, depending on source.
Evidence provenance	Which adapter/source and feature window supports each displayed contributor.

10 Attribution, explainability and contributor semantics

10.1 Four contributor categories

Category	Evidence family	Output wording
Physical exertion strain	Movement intensity/activity bouts, workout overlap/recency, HR aligned to movement.	“Activity-associated elevation” or “movement/workout evidence coincides with the change.”
Non-exertional physiological elevation	HR or other valid physiological feature above personal reference while measured activity/recent workout context is low.	“Non-exertional physiological elevation observed.” Do not name psychological stress unless a suitable self-report association is explicitly available.
Recovery deficit / accumulated load context	Nightly HRV, sleep, resting HR, activity/load deviations across eligible history.	“Recovery-context deviation” or “sleep/HRV/resting-HR pattern differs from your baseline.”
Systemic / respiratory / vital trend	Available SpO2, respiratory rate or temperature deviations.	“Respiratory/oxygen/vital trend differs from your reference; verify measurement and observe trend.” Never diagnose.

10.2 SHAP and tree attribution

TreeSHAP may be used only with the exact compatible tree model, feature order/schema and inference version it explains. SHAP values describe how model features contribute to the model’s prediction relative to its explanation baseline. They do not prove a biological mechanism or identify an external cause.

If displaying shares of attribution, label them “relative share of positive model evidence” under stated normalization and only when all compared components are valid. Do not call them “probability this cause produced strain” or force 100% attribution shares when contributors can co-occur or evidence is incomplete.

10.3 Evidence record shape

```
ContributorEvidence {
  category: activity | non_exertional | recovery | vital_trend
  evidence_type: measured_deviation | model_association | policy_trigger
  feature_name: versioned_feature_id
  observed_value / personal_reference / robust_deviation
  direction: above | below | mixed | unavailable
  evidence_start_utc / evidence_end_utc
  source_ids: [provenance references]
  quality / coverage / freshness
  model_id + model_version: nullable
  attribution_method: rule | TreeSHAP | other validated method
  causal_claim: false
}
```

10.4 Example of defensible explanation

> Good output
> “Heart rate was above your recent personal reference during a low-movement window. The connected source provided no raw acceleration for this period, so the activity context is based on steps and workout records and has lower temporal resolution. This is a physiological pattern, not proof of a specific cause.”

Avoid explanations such as “work stress caused your heart rate to rise” unless that exact context was measured in a suitable study—and even then the result should be described as an association rather than a causal determination.

11 Recovery-context engine and recommendations

11.1 V1 recovery context

Recovery context is a personalized summary constructed from available valid nightly HRV, sleep, resting-HR and recent activity/load features relative to the user's baseline. It is not the body's actual recovery state, a clinical readiness assessment, or a universal measure comparable across devices.

V1 should prioritize transparent descriptive outputs: which signals deviated, whether they are fresh, what historical comparison was available, and how the pattern changed over prior valid observations. If the user has only nightly summaries, output daily context—not fictitious minute-level kinetics.

11.2 Recovery trajectory modeling gate

- Define an eligible event: onset, event end, evidence window, pre-event baseline and endpoint of recovery before training a predictor.
- Define return-to-baseline mathematically with an explicit tolerance band and minimum dwell time; do not select the endpoint after seeing model predictions.
- Require enough valid episodes from multiple participants and separate participants/episodes for holdout evaluation.
- Evaluate time-to-endpoint error (for example MAE), interval coverage and failure rate; report uncertainty and per-person spread.
- Use a simple exponential/quantile curve only if it fits the observed data better than transparent baselines. Otherwise show the observed trajectory only.
- Do not keep legacy fixed recovery-time heuristics (for example tau values of 6.5/14 minutes or generic 8-12 hour claims) unless the exact endpoint and held-out evaluation support them.
- Disable "minutes to recovery" if event episodes, labels or prediction interval calibration are insufficient.

11.3 Deterministic recommendation policy

Evidence pattern	Conservative general-wellness guidance	Guardrail
Activity/workout evidence	Cool down, rest as appropriate, hydrate according to personal needs, and review activity context.	No medical certainty; do not prescribe an exact recovery duration without evidence.
Non-exertional physiological elevation	Take a brief pause, consider a gentle paced-breathing break, and check context/measurement quality.	Do not label anxiety or psychological stress from sensor values alone.
Recovery-context deviation	Consider sleep opportunity, rest and manageable activity; review recent load and trends.	Do not diagnose overtraining or guarantee readiness.
Unusual SpO2/respiration/temperature trend	Check fit and repeat according to device guidance; consider appropriate professional advice if persistent or concerning.	No diagnosis; account for device error, context and source validity.
Insufficient data / conflicting evidence	Explain which signal or history is missing; suggest checking wearable fit/sync or waiting for a valid observation.	Never offer false reassurance or fabricate a primary contributor.

11.4 Optional LLM narration

The LLM is a language layer, not the sensor model, score calculator, medical decision-maker or cause-attribution engine. It receives a schema-validated evidence bundle and an approved policy result. It may simplify language but may not invent missing features, alter scores, introduce diagnosis, manufacture probabilities or override safety policy.

Allow-listed narration input: score name/value/version and calibration status; exact state semantics and evidence windows; contributor records with quality/coverage; recovery context and observed trend; policy-approved recommendations and IDs; missing/stale signals and temporal resolution.

- Every narration statement must be traceable to the supplied evidence or an approved policy.
- No causal wording, medical diagnosis, unsupported number/time estimate or fabricated signal.
- Uncertainty, missing data and coarse temporal resolution must be retained when relevant.

12 Datasets, evidence and training plan

Dataset choice should be driven by the target a model is supposed to learn. No one dataset provides every universal wearable signal, a true personalized baseline, a real-world stress label, and a ground-truth recovery endpoint at once. Keep each dataset's labels and time semantics explicit; combine modules, not incompatible label meanings.

12.1 Core longitudinal dataset roles

Dataset	Signals / labels (reported in source)	Fit for the capstone	Cautions
Wearable HRV + sleep-diary dataset (Scientific Data, 2025)	49 participants over four weeks; wearable HRV/PPG and motion-related data; five-minute HRV; daily sleep diaries; biweekly PHQ-9/GAD-7/ISI questionnaires.	Personal baseline/history feature engineering; real-world HR/HRV/activity quality; relationship between wearable summaries, sleep diary and longer-term questionnaires.	Questionnaires collected biweekly are not minute-level labels. Do not claim EEG sleep-stage ground truth from sleep diaries. Verify local source/license/schema.
RELAX six-week longitudinal stress dataset (Scientific Data, 2026)	31 working adults; six weeks; timestamped IBI, 52-Hz tri-axis ACC, repeated self-report stress assessments.	Within-person longitudinal context; HR/IBI + movement; self-report association target; baseline and activity-confounding evaluation.	A self-report is not objective physiological truth; report time may not be exact onset. Validate schemas, access terms and participant-disjoint splits.
Neurophysiological stress-resilience during HCI (PhysioNet)	Lab task phases such as rest, baseline, stress and recovery; physiological and additional modalities.	Optional acute task-associated state or recovery-episode evaluation when the actual released fields match product requirements.	Lab task phases are not interchangeable with real-world self-reported stress; this is an optional supporting source, not required for V1.
PAMAP2 physical activity	Activity labels and multiple IMUs, including body-worn inertial signals.	Optional activity/exertion-supervision dataset.	Activity classification transfer needs held-out participants and target-source/domain validation; not stress supervision.
BIDSleep / multi-night HR + ACC with EEG sleep labels	47 participants and 253 nights; Apple Watch HR and three-axis ACC with EEG-derived stage labels.	Optional if the product truly needs a sleep-stage fallback and has compatible inputs.	Not direct momentary stress labels or a general four-week stress baseline; use only for a defined gap.

12.2 Source/data acceptance checklist

- Confirm the primary publisher/repository, dataset version, license/access terms and exact local files before training.
- Create a provenance manifest with subject identifiers, sessions, channels, units, sampling rates, timestamps, labels, exclusions and checksums.
- Audit label semantics and collection timing; do not treat retrospective questionnaires as exact moment-level ground truth.
- Create deterministic sample/window counts with subject/session leakage checks and a reproducible preprocessing manifest.
- Keep target/source restrictions in reports. A model trained on self-report association cannot be renamed an objective stress detector.

12.3 Dataset roles and candidate heads

The broader research architecture may evaluate activity supervision, acute task-state estimation, self-report association, vital-trend deviation and recovery trajectory as distinct heads. Dataset candidates such as NeuroStress, PAMAP2 and BIDSleep are only included when a concrete gap is demonstrated and their exact accessible schemas are verified. They are not all mandatory dependencies for the first release.

The main state estimator should start with robust personalized deviations and transparent rules. Train one calibrated XGBoost/LightGBM association model only when the available labels and participant-disjoint data volume support the defined target. If labels are sparse or incompatible, do not fabricate a trained classifier;

retain the measured/heuristic outputs and label them accurately.

12.4 Do not use these as universal evidence

- Legacy WESAD heads trained on BVP/EDA/TEMP windows; the old representation contract is not compatible with universal HR/HRV/sleep input.
- Unverified informal dataset titles or secondary catalog pages without an authenticated primary source, release version, schema and license.
- The historical WESAD metrics as proof of universal multi-brand performance; re-run them only if discussing the legacy work, and clearly mark them as legacy results.
- Session-relative timestamps as calendar dates or synthetic multi-day histories without actual timestamps.

13 Training, validation and scientific defensibility

13.1 Leakage-safe splits

- Use participant-disjoint train/validation/test splits for all supervised results when participants are the generalization unit.
- Where longitudinal forecasting/trajectory is involved, preserve temporal order and evaluate future episodes without training on future windows from the same target episode.
- Fit normalization, imputers (where scientifically permitted), PCA, feature selection, class balancing, calibration and thresholds on training/validation only as appropriate; never fit these on the test set.
- Keep all windows from an episode and any duplicated/reconstructed sample in the same split. Document participant, session and device/source overlap.
- Run cross-source/leave-one-device-out validation only when enough compatible data exist; otherwise state that universal generalization remains unproven.

13.2 Metrics by task

Task	Primary metrics	Must also report
Activity/exertion	Macro F1, balanced accuracy, confusion matrix.	Per-participant results; class support; device/source transfer; feature ablation.
Self-report association / state	AUROC and AUPRC where suitable; Brier score; calibration curve.	Label timing window; per-participant metrics; sensitivity/specificity at chosen operating point; prevalence.
Personal baseline	Deviation stability, missingness/coverage behavior, false-alert burden under defined test conditions.	Global-vs-personal comparison; minimum-history sensitivity; false alerts by participant/device.
Forecasting	MAE/quantile loss vs simple baselines.	Prediction-interval coverage; horizon-specific errors; no label leakage.
Recovery trajectory	Time-to-endpoint MAE and interval coverage; failure/no-endpoint rate.	Episode eligibility, endpoint definition, participant-held-out results and uncertainty.
Universal/adapter robustness	Schema validity; supported-signal coverage; failure and fallback rates.	Per-source capability matrix; quality degradation; missing-modality ablations.

13.3 Calibration and uncertainty

If the classifier emits probabilities, calibrate on a validation set that is separate from final testing and report reliability curves and Brier score. Probabilities refer to the exact trained target (for example “association with a time-aligned self-report”), not a latent truth. If a head is out of domain, the target modality is missing or the baseline is uncalibrated, suppress/mark its probability rather than forcing a confident number.

13.4 Required ablation studies

Comparison	Question answered
Global thresholds vs personal robust baseline	Does personalization improve useful detection or reduce false alerts across individuals?
HR-only vs HR + activity context vs full available context	How much does movement/context reduce confusion between exertion-associated and non-exertional elevation?
ENMO/statistical ACC vs NormWear embedding vs fused	Does the frozen encoder add value beyond simpler movement features on held-out participants?
Rules/robust deviation vs calibrated supervised model	Does machine learning improve the named target without harming calibration or interpretability?
Full signals vs simulated missing-modality profiles	Does capability-aware behavior degrade gracefully when brands expose different channels?
Observed recovery trend vs predicted duration	Is a trained trajectory model accurate and calibrated enough to justify showing a time estimate?

13.5 Acceptance thresholds are predeclared

Do not choose success thresholds after seeing test results. Before evaluation, define minimum coverage, minimum participant/episode counts, calibration tolerances, false-alert budget, acceptable confidence-interval width and per-source generalization tests. If sample size does not support strong conclusions, document descriptive results and limits rather than claiming validation.

14 Strain/recovery outputs and API contract

14.1 Stable output semantics

Field	Purpose / required semantics
physiological_deviation_score	Optional 0-100 internal, versioned wellness index. Not a clinical unit and not directly comparable across devices without calibration.
state_association	A target-specific state/association such as activity-associated elevation, non-exertional physiological elevation, recovery-context deviation, respiratory/oxygen deviation, or insufficient data. Probabilities only when a head has been trained and evaluated for that exact label.
contributors[]	Structured feature evidence: current value, personal reference, deviation/direction, evidence window, source, quality and classification as observation vs contextual association.
recovery_context	Available HRV/sleep/resting-HR/load deviations, baseline sufficiency and observed trend. Predicted minutes-to-recovery disabled until independently validated.
recommendations[]	Deterministic policy ID/version, trigger evidence and approved general-wellness suggestion. Narration cannot change the evidence.
data_quality	Per-signal availability, freshness, coverage, missing reason and maximum supported temporal resolution.

14.2 Example JSON response


```
{
  "schema_version": "1.0",
  "generated_at_utc": "2026-10-15T12:00:00Z",
  "physiological_deviation": {
    "score": null,
    "status": "experimental_or_unavailable",
    "baseline_status": "provisional",
    "interpretation": "personal_deviation_not_clinical"
  },
  "state_association": {
    "label": "non_exertional_physiological_elevation",
    "probability": null,
    "method": "rule_based",
    "target_semantics": "physiological_pattern_not_psychological_diagnosis"
  },
  "contributors": [{
    "category": "non_exertional",
    "evidence_type": "measured_deviation",
    "feature_name": "heart_rate_vs_personal_reference",
    "direction": "above",
    "evidence_window_minutes": 15,
    "source_quality": "moderate",
    "coverage": 0.82,
    "causal_claim": false
  }],
  "recovery_context": {"status": "insufficient_history", "predicted_minutes": null},
  "recommendations": [],
  "data_quality": {
    "raw_acc_available": false,
    "activity_proxy_available": true,
    "hrv_status": "nightly_summary_only",
    "max_supported_resolution": "window_or_daily",
    "missing_signals": ["raw_acc"]
  }
}
```


The JSON is an illustrative contract, not a claim that these example values came from a real participant. The production API must include actual timestamps, units, feature-window IDs, source provenance and numerical coverage semantics defined in the typed schema.

14.3 API behavior

- Reject invalid payloads with typed errors rather than silently changing data meanings.
- Keep model version, feature schema, baseline version, policy version and adapter version in the response or retrievable provenance bundle.
- Return null/unavailable where no supported value exists. Never place a fabricated zero in an unavailable field.
- Allow the UI to explain why the estimate is coarse or unavailable: no raw ACC, no valid IBI, stale nightly summary, insufficient baseline, low coverage or unsupported device capability.

15 Reference repository structure

The repository should reflect data contracts, feature engineering, modeling, policy and product UI as separate responsibilities. Adapt names to the actual repository after auditing existing code; do not copy an entire legacy source tree into the clean product.

```
universal-wearable-engine/
|-- apps/web/                        # universal product UI
|-- services/{api,worker}/
|-- src/universal_engine/
|   |-- domain/{signal_contracts,device_capabilities,feature_schema,output_schema}.py
|   |-- adapters/{base,csv_health_export,wearable_hrv_sleep_dataset,relax}.py
|   |-- ingestion/{validator,timestamp_alignment,deduplication}.py
|   |-- quality/{signal_quality,coverage,freshness}.py
|   |-- preprocessing/{accelerometer,heart_rate,hrv_ibi,sleep,daily_context}.py
|   |-- features/{kinematic,physiological,temporal,context}_features.py
|   |-- baselines/{personal_baseline,robust_scaling,baseline_update_policy}.py
|   |-- models/{activity_classifier,state_association,vital_trend_detector}.py
|   |-- models/recovery_trajectory.py   # disabled unless validated
|   |-- fusion/{contributor_fusion,deviation_index,confidence}.py
|   |-- explainability/feature_attribution.py
|   |-- recommendations/{policy_engine,safety_guardrails,llm_narrator}.py
|   |-- aggregation/{event_aggregation,daily_aggregation}.py
|   `-- orchestrator.py
|-- configs/{signal_schema,device_capabilities,feature_contracts,model_registry}.yaml
|-- scripts/{audit_datasets,train_activity_model,train_state_model,evaluate_models}.py
|-- tests/{unit,integration,leakage,model_contracts}/
|-- reports/{dataset_audit,model_evaluation,ablation_study}.md
|-- pyproject.toml
`-- README.md
```

15.1 Module boundaries

Component	Must own	Must not own
Adapter	Vendor/export-specific auth, field mapping, units and capability declaration.	Universal scoring logic or assumptions about every other device.
Contract/quality	Canonical schema, validation, freshness and coverage rules.	Learning target or polished user-facing interpretation.
Features/baseline	Deterministic feature calculations and causal personal history.	Unversioned model-specific feature order or future observations.
Model	Task-specific prediction for one defined label, train/inference schema and calibration.	Inventing missing data or deciding medical recommendations.
Fusion/explanation	Combine supported evidence, uncertainty and feature/model explanations.	Claiming causality from correlation or normalizing unsupported values.
Policy/narration	Approved action mapping and optional plain-language wording.	Recomputing scores, inventing causes or altering state semantics.

15.2 Model and artifact management

- Store large pretrained checkpoints in an artifact store or deliberate managed location rather than blindly committing them to ordinary Git history.
- Keep checksums, source URL/commit, model license, preprocessing version, feature schema, training dataset manifest and evaluation report with each model.
- For any NormWear experiment, pin the official repository commit, checkpoint hash, CWT/config details and supported input contract.
- Do not let the main product import WESAD-specific pipeline code or legacy trained classifier heads at runtime.
- Every feature/head in docs and UI should be tagged implemented, experimental, planned or unsupported.

16 Implementation roadmap and acceptance gates

16.1 Phased delivery plan

Phase	Work	Exit gate
0. Audit and protect existing work	Inventory repository, source paths, checkpoints, datasets, labels, current Git state and uncommitted changes. Preserve legacy artifacts if reproducibility is required.	Source inventory and dataset audit written; no destructive history rewrite; known assets and unresolved gaps documented.
1. Universal contracts	Define canonical observation/output schemas, device capability registry, versioning and contract tests.	Synthetic and real sample payloads validate; missing data has explicit semantics; no legacy-specific runtime dependency.
2. Real data adapters	Implement actual adapters for the selected verified longitudinal HRV/sleep dataset and RELAX; build manifests and coverage reports.	Signals, units, timestamps, labels, licenses/access and subject counts are verified locally; deterministic loading passes.
3. Feature + baseline engine	Implement HR, valid HRV/IBI, activity, sleep/context features; robust causal personal baseline; missingness and workout eligibility.	No future leakage; MAD=0 and sparse history handled; global vs personal baseline comparison report exists.
4. State/contributor model	Train one target-specific calibrated tree model only if labels and splits support it; add activity/context gating and explanatory evidence.	Participant-held-out metrics, calibration, per-participant analysis and data lineage are reported.
5. Recovery + policy	Show recovery-context and observed trend; define event endpoints; add deterministic recommendation policies; optional LLM narration.	No recovery-duration prediction before held-out validation; all recommendations trace to policy/evidence.
6. One real device integration	Select a feasible, accessible device/API path and complete an end-to-end flow. Expand adapters only after verified access.	At least one actual device path tested; capability matrix and permissions documented; no empty adapter stubs presented as support.
7. UI and release audit	Expose score semantics, contributors, baseline status, temporal resolution, missingness and provenance. Run ablations and final tests.	Claim-to-evidence review passes; docs accurately mark supported/experimental/planned/unsupported features.

16.2 Test matrix

Test group	Required cases
Robust math	MAD=0; near-zero scale; outliers; empty history; only one eligible observation; clipping and fallback.
Missingness/capability	No raw ACC; no HRV; only nightly HRV; no sleep stages; optional SpO2 absent; unsupported signal and denied permission.
Time	Timezone/daylight-saving boundary; midnight roll-over; out-of-order samples; duplicates; stale summary; relative session time.
Quality/alignment	Low coverage; artifact-tagged IBI; HR/ACC windows with no overlap; mixed cadence; inconsistent units.
Baseline leakage	Scores cannot use future values; baseline updates exclude ineligible workouts/bad data; training and test participants do not overlap.
API/model contracts	Unknown schema version; feature ordering change; unsupported model input; missing required feature; model/calibration version mismatch.
Policy safety	No diagnosis; no unsupported causal language; no fabricated score/probability; correct insufficiency message; LLM cannot override structured output.

16.3 Definition of done

- The product’s data flow works end-to-end for at least one verified source, with all missing signals and assumptions visible.
- Every model has a named target, source and feature manifest, leakage-safe evaluation, calibrated outputs where applicable, and declared limits.
- Personal baseline behavior is causal, quality-aware and tested for sparse/constant/outlier histories.

- Attribution output differentiates measured deviation from model association and contextual interpretation.
- Recovery is descriptive until episode-level validation supports prediction; recommendations are rule-guarded.
- The final UI and capstone report do not claim 100% vendor compatibility, medical validity or causal stress attribution without evidence.

17 Risks, limitations and decision log

17.1 Key risk register

Risk	Why it matters	Mitigation / decision
Label mismatch	Lab phase, self-report, activity class and physiological signal are not the same target.	Separate heads and target names; verify timing/semantics; do not build a blended universal “stress” label.
Device access mismatch	Consumer app features may not be available through public APIs or on every device model/OS.	Start with one verified adapter; maintain capability matrix; document raw vs export vs summary access.
Poor personal history	Sparse/stale observations make robust deviation unstable.	Minimum sample/coverage gates; provisional/unavailable state; confidence is explicit.
Cross-person leakage	Overlapping windows can inflate metrics and falsely imply generalization.	Participant-disjoint splits; episode grouping; deterministic manifests and leakage tests.
Model complexity without evidence	Foundation models or temporal nets may add cost without improving the product task.	Transparent baselines first; add model only after task-level ablation improves held-out results.
False causal language	Sensor correlation does not identify a unique real-world cause.	Use association language; separate measured contributor from possible context; avoid exact causal shares.
Unsupported recovery estimates	Recovery endpoint varies and may be unobserved.	Observed trajectories only until explicit endpoints/episodes and calibrated held-out evaluation exist.
Clinical/safety overreach	Users can overinterpret scores or vital deviations.	General-wellness framing, cautious policy, measurement recheck and appropriate escalation language for persistent/concerning readings.
Schema drift	Provider API/SDK changes alter units, cadence or access.	Version adapters/schema; contract tests; monitor breakage; preserve raw source metadata and manifests.

17.2 Important implementation decisions

- Keep one product and one canonical pipeline. Do not retain a parallel active WESAD product track.
- Preserve old WESAD work only as an archive/legacy snapshot if reproducibility is useful; do not import its classifiers or assumptions into the live product.
- Reuse robust baseline ideas only after code review and tests; replace zero-MAD handling, fixed weights and unvalidated recovery rules.
- NormWear is a candidate frozen ACC representation, not a mandatory dependency and not an activity classifier. Its downstream value must be proven against ENMO/statistical features.
- Chronos-Bolt/TCN, additional foundation models and LLM decision logic are not necessary for V1. Forecasting remains optional and task-specific.
- The first release can be valuable with transparent baselines and rules if a suitable supervised target is not supported by the available labels.

17.3 Language guide

Prefer	Avoid
Personal physiological deviation	Universal “normal/abnormal” claims with no calibration.
Activity-associated elevation	Activity definitely caused the elevation.
Non-exertional physiological elevation	Psychological stress detected from HR alone.
Recovery-context deviation / observed trajectory	True recovery readiness or exact minutes-to-recovery without validated data.
Association with contemporaneous self-report	Objective stress ground truth when the label is a self-report.
Relative share of positive model evidence	Exact causal percentage or probability of cause.
Insufficient data / ambiguous context	Normal, healthy or safe when the relevant signal is missing.

18 References and technical resources

The following primary or official resources are relevant to the dataset and technical decisions summarized in this specification. Verify the exact version, access terms, and schema during implementation; a citation does not imply the dataset has already been downloaded or audited.

[1] Baigutanova et al. (2025). “A continuous real-world dataset comprising wearable-based heart rate variability alongside sleep diaries.” Scientific Data 12, 1474. Dataset context: longitudinal wearable HRV, movement-related records, sleep diaries and biweekly assessments.
https://www.nature.com/articles/s41597-025-05801-3

[2] Dataset DOI for the 2025 wearable HRV + sleep-diary dataset.
https://doi.org/10.6084/m9.figshare.28509740

[3] Halmich et al. (2026). “A six-week longitudinal dataset of wearable and self-reported stress measurements in working adults.” Scientific Data. RELAX dataset context: IBI, tri-axis ACC and repeated self-report assessments.
https://www.nature.com/articles/s41597-026-07711-4

[4] RELAX data DOI.
https://doi.org/10.5281/zenodo.18693288

[5] Shaffer, F. & Ginsberg, J.P. (2017). “An Overview of Heart Rate Variability Metrics and Norms.” Frontiers in Public Health, 5:258. HRV metrics and interpretation.
https://doi.org/10.3389/fpubh.2017.00258

[6] Official NormWear repository. Use for source, code/version, checkpoint and supported input audit before optional ACC integration.
https://github.com/Mobile-Sensing-and-UbiComp-Laboratory/NormWear

[7] PhysioNet: Neurophysiological Dataset of Stress Resilience During Human-Computer Interaction. Optional acute task/recovery research source.
https://physionet.org/content/neuro-stress-resilience-hci/1.0.0/

[8] UCI Machine Learning Repository: PAMAP2 Physical Activity Monitoring. Optional activity-supervision source.
https://archive.ics.uci.edu/dataset/231/pamap2%2Bphysical%2Bactivity%2Bmonitoring

[9] PhysioNet: A Multi-Night Instantaneous Heart Rate and Accelerometry Dataset with EEG Sleep Stage Labels (BIDSleep). Optional sleep-stage source.
https://physionet.org/content/bidsleep-dataset/1.0.1/

[10] Samsung Health Sensor SDK data specifications. Check model/OS/SDK-specific signal access and cadence.
https://developer.samsung.com/health/sensor/guide/data-specifications.html

[11] Google Health API access and migration documentation. Reconfirm current onboarding/access before selecting an integration path.
https://developers.google.com/health/about

[12] Google Health data-access migration documentation.
https://developers.google.com/health/migration/data-access

A1 Glossary

Term	Definition in this project
ACC	Accelerometer; here, raw tri-axis movement samples when actually available.
ENMO	Euclidean Norm Minus One, a movement-intensity feature derived from acceleration magnitude and gravity handling.
HR	Heart rate.
HRV	Heart-rate variability; keep metric and measurement window explicit.
IBI / NN interval	Inter-beat interval; NN conventionally refers to intervals between accepted normal beats after artifact handling.
MAD	Median absolute deviation, a robust dispersion estimate used for personal baselines.
Personal baseline	Causal, quality-filtered summary of the user’s prior measurements in an appropriate context.
Contributor evidence	Measured deviation or model association that provides support for a category; not automatically a cause.
State association	A label tied to an explicit target such as activity-associated elevation or contemporaneous self-report association.
Calibration	Assessment/adjustment of model probabilities so stated confidence matches observed frequency for the target.
Coverage	Amount of expected signal/time window represented by valid observations.
Provenance	Traceable source, adapter, version, timestamps, processing steps and model lineage for an observation or output.
Observed recovery trajectory	A measured pattern across valid windows after a defined event, without a prediction implied.
Capability-aware	Behavior adapts to the signals and temporal fidelity actually available from the selected source/device.

A2 Pre-release checklist

Use this compact checklist as a final technical and academic review before the capstone demonstration.

- [ ] Only real, permitted, documented signals enter the engine; missing modalities remain explicitly missing.
- [ ] The exact data source, version, license/access and label semantics are recorded.
- [ ] Timestamps, timezone, units, sampling rates, intervals and provenance are preserved.
- [ ] Personal baseline uses only prior eligible data and handles zero MAD, sparse history, bad coverage and staleness.
- [ ] HRV is device-reported or calculated from valid timestamped IBI/NN data, never inferred from smoothed BPM.
- [ ] Activity proxies are not mislabeled as raw ACC, ENMO or NormWear embeddings.
- [ ] Each supervised head has a defined target and participant-disjoint evaluation; calibration is reported for probabilities.
- [ ] TreeSHAP or other explanations are described as model evidence, not causal attribution.
- [ ] Recovery predictions remain disabled until valid episodes/endpoints and held-out uncertainty checks pass.
- [ ] All guidance traces to a versioned policy and does not diagnose or invent external causes.
- [ ] UI shows data quality, baseline status, contributor evidence, temporal resolution and missing signals.
- [ ] Docs clearly label each feature implemented, experimental, planned or unsupported.

Final architectural principle
A trustworthy universal wearable engine does not force every device into the same sensor model. It provides one consistent interpretation framework while preserving source fidelity, explicit gaps, personal history, uncertainty and evidence provenance.